

PROBLEMSHEET-3
EPMTC301: MATHEMATICS FOR MACHINE LEARNING
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Here, we follow the same notations used in class. If you find any typos please let me know ASAP. It is highly recommended to try the first problem.

- (1) Do the following problems from the book **Schaum Series**:
- (a) Page number 258-262, Exercise number: 7.57, 7.61, 7.64, 7.65, 7.66, 7.70, 7.75, 7.76, 7.77, 7.78, 7.87, 7.88.
 - (b) Page number 321-323, Exercise number: 9.45, 9.48, 9.49.
- Link for book: [click here](#)

- (2) Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear map defined by $T(x, y) = (-3y, x)$. Find the eigenvalues of T .
- (3) Let $T : P_4(\mathbb{R}) \rightarrow P_4(\mathbb{R})$ be a linear map given by $T(f(x)) = x \frac{d}{dx}(f(x))$. Find all eigenvalues and eigenvector of T .
- (4) Let $T : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ be a linear map given by $T(f(x)) = \frac{d}{dx}(f(x))$. Find all eigenvalues and eigenvector of T .
- (5) Let T be an invertible linear map on vector space V with eigenvalues $\lambda_1, \dots, \lambda_n$ and eigenvectors $v_1, \dots, v_n$. Find the eigenvalues and eigenvectors of T^{-1} .
- (6) Let $V = P_2(\mathbb{R})$ be the space of real polynomials of degree less or equal 2 with the following inner product

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x)dx, \quad f, g \in P_2(\mathbb{R}).$$

Find an orthonormal basis of $P_2(\mathbb{R})$.

- (7) Let T be the linear operator on $\mathbb{R}^3$ defined by

$$T(x_1, x_2, x_3) = (3x_1, x_1 - x_2, 2x_1 + x_2 + x_3).$$

Find all eigenvalues and eigenvectors of T .

- (8) Prove that the matrix $\begin{pmatrix} 1/\sqrt{5} & 2/\sqrt{5} \\ 2/\sqrt{5} & -1/\sqrt{5} \end{pmatrix}$ is an orthogonal matrix and find its eigenvalues.